Weighted Weil-restriction decompositions

2026-10-19

Native larger-curve diagnostic

This run uses the repository's native cubic-field arithmetic and S3 resultant pair solver. It constructs no covering curve and no whole-field or whole-group operation table. Each decision is cross-checked with a separate signed-pair dictionary unavailable to the solver.

The local reference is the identical hybrid with c=1, not the best complete IC pipeline. Counts below are stage diagnostics, not end-to-end speedups. Complete S, ratios to rho and external fastest-tool comparisons are unestablished. Wall time is not admitted at L0.

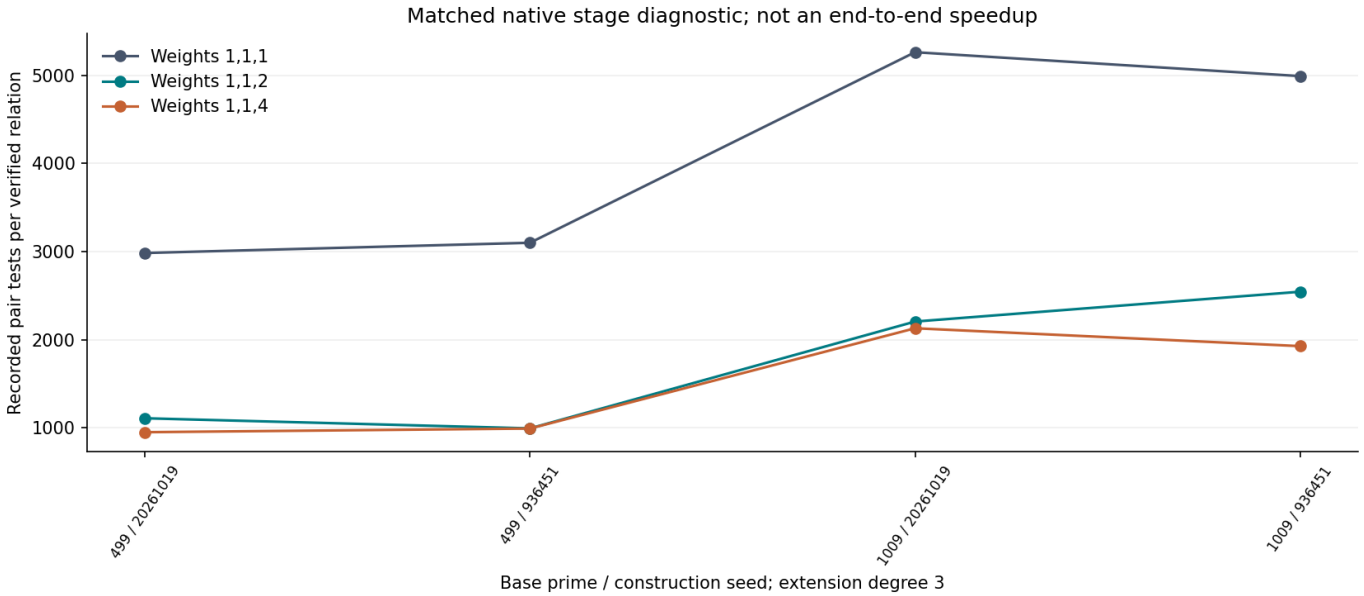


Figure 1: Recorded pair queries per verified relation; native-20261019-followup

p	construction seed	group order	unknowns	weight	verified / targets	pair tests	pair tests / relation
499	20261019	124249159	249	1	19 / 128	56670	2982.632
499	20261019	124249159	249	2	47 / 128	52049	1107.426
499	20261019	124249159	249	4	52 / 128	49357	949.173
499	936451	124261523	258	1	19 / 128	58878	3098.842
499	936451	124261523	258	2	51 / 128	50621	992.569
499	936451	124261523	258	4	52 / 128	51520	990.769
1009	20261019	1027186051	495	1	21 / 128	110487	5261.286
1009	20261019	1027186051	495	2	45 / 128	99211	2204.689
1009	20261019	1027186051	495	4	47 / 128	100052	2128.766
1009	936451	1027286801	495	1	22 / 128	109779	4989.955
1009	936451	1027286801	495	2	42 / 128	106812	2543.143
1009	936451	1027286801	495	4	52 / 128	100150	1925.962

p	construction seed	group order	unknowns	weight	verified / targets	pair tests	pair tests / relation

Synthetic one-target recoveries

p	seed	weight	residuals	independent rank	status
499	20261019	1	1579	228	verified_recovery
499	20261019	2	590	227	verified_recovery
499	20261019	4	565	229	verified_recovery
499	936451	1	1430	244	verified_recovery
499	936451	2	590	247	verified_recovery
499	936451	4	557	241	verified_recovery
1009	20261019	1	3044	453	verified_recovery
1009	20261019	2	1247	445	verified_recovery
1009	20261019	4	1152	443	verified_recovery
1009	936451	1	3063	448	verified_recovery
1009	936451	2	1302	470	verified_recovery
1009	936451	4	1177	457	verified_recovery

Completed 12 recoveries; 0 budget-exhausted recoveries. Total independently checked PDP decisions, including A/A controls and recovery attempts: 18344. The independent comparison is against a separate dictionary, not an independent host.

Limits and next checks

The legacy pair API does not emit internal lookup, polynomial addition/division, allocation or RNG counts. Its F_p multiplication counter also uses legacy inverse accounting. These missing counts prevent a complete pricing band, S, or ranking against rho. No strongest-external-solver comparison was run: Magma, msolve and FGb were not run here. Base and weighted-slot setup, verification group operations, residual-generation group operations and modular elimination multiplications have separate ledgers. Dictionary construction, reference arithmetic and fixture generation are excluded validation/setup work without complete operation ledgers. No binary-field or ECC2K transfer was tested.

The next discriminating step is complete native counters plus matched ecbench one-target rho measurements, with independent-host replay and the fastest available external algebraic reference. These obligations remain open; verified recoveries alone do not discharge them.

Reproduction and provenance

See config.json, host.txt, SHA256SUMS, records.jsonl, the committed PROTOCOL.md and the native example. Seed 936451 is the holdout. Inputs and outputs are public synthetic fixtures. Histories from the earlier Python experiment remain historical and are not runtime baselines. All A/A decisions and deterministic counters were asserted equal.

Follow-up run record

Run: native-20261019-followup; frozen source/protocol 040b8e1a. All four fixtures, 16 PDP arms (128 targets each) and 12 recoveries completed. PARI confirmed all fixtures. The 18,344 dictionary checks include repeated primary stage panels, not independent samples. This exploratory extension adds full recoveries on the same four larger curves. No uncertainty interval or runtime ratio was computed. Host: AMD EPYC 9V74 KVM, four-CPU quota, 16 GiB, L0 concurrent. Exact command and budget: config.json. Raw records, provenance, binary hash, PARI checks, and source snapshot retain the evidence.